12 A radioactive substance undergoes decay. Its mass, M grams, after t days is defined by $M = ae^{-kt}$, where a and k are constants.

The table below shows some measurements of M and t.

t (days)	1	4	7	10	12
M (grams)	1.40	1.06	0.82	0.63	0.53

(a) Plot $\ln M$ against t and draw a straight line graph to illustrate the information. [2]

(b) Use your graph to estimate the value of a and of k. [4]

(c) Use your graph to estimate the time taken, to the nearest day, when at least 60% of the radioactive substance has decayed. [2]